

Supplementary Material

Serving the Public as Civil Servants or Citizens? Public Employees' Exposure to Public Participation, Job Satisfaction, and Volunteering

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This supplementary material includes:

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S1. Survey Sampling, Administration, and Weighting

This cross-sectional survey, conducted in 2018, represents the fifth wave of a multi-year research project on public managers in local governments. Previous waves were conducted in 2010, 2012, 2014, and 2016. The initial sampling frame was developed in 2010 based on the city's population and has been updated in subsequent waves. For the 2018 survey, the research team used this existing sampling frame, incorporating updates from the prior waves. In 2018, the research team surveyed volunteering behaviors for the first time.

In spring 2018, the research team reviewed city department websites to verify whether the public managers who had been sampled in earlier surveys were still serving in the same positions. They updated contact and demographic information for individuals who had changed positions and confirmed existing information for those who remained. When information was not available online, the researchers contacted municipal offices directly to confirm the information of department directors across five departments.

The distribution of cities in the sampling frame is heavily skewed toward smaller cities: 84% are small-sized cities (population 25,000–100,000), while only 16% are medium-sized cities (population 100,000–250,000). Given that larger cities typically have more resources and capacity to support citizen engagement and that residents in larger cities are generally more likely to engage in civic activities, the research team employed a stratified probability sampling strategy to develop sample frame. The research team selected a census of all medium-sized cities ($n = 184$) and drew a proportional sample of 316 small-sized cities.

Among the small-sized cities, the proportional sample included:

- 59% from cities with populations between 25,000 and 50,000,
- 28% from cities with populations between 50,000 and 75,000, and
- 13% from cities with populations between 75,000 and 100,000.

Table S1.1 below presents a detailed breakdown of the sampling strategy used to select the 500 cities included in this study.

Table S1.1. Sampling Strategy

City types	City population	Number of cities	Proportion over all cities	Proportion in small-size cities strata*	Number of sampling cities	Proportion of sampling small-size cities in strata	Total sample
Small-size (25k-100k)	25-50k	591	50%	59%	186	59%	186
	50-75k	278	23%	28%	88	28%	88
	75-100k	133	11%	13%	42	13%	42
Medium-size (100k-250k)	100k-125k	68	6%				68
	125-150k	37	3%				37
	150-175k	23	2%				23
	175-200k	28	2%				28
	200-225k	18	2%				18
	225k-250k	10	1%				10
Totals		1186	100%	100%	316	100%	500

*There are 1002 small-size cities in total.

The survey was administered online using Sawtooth Software®. Before sending survey invitation to public managers through email, the research team verified the validity of email addresses with PING. A pre-notification email was sent on April 6th, 2018 to notify sampled public managers that they would soon receive a survey invitation. An email invitation (including unique ID, passwords, and hyperlink to the survey) was sent on April 18th, 2018. To boost response rate, six reminder emails were sent at ten-day intervals between April 30th to June 27th, 2018. Additionally, follow-up phone calls were made to nonrespondents between the fourth and final reminders. The survey was closed on August 7th, 2018. After removing invalid email addresses (i.e., undeliverable emails during PING check and email communication) and ineligible managers (i.e., those who retired or left their positions), the final valid sample consist of 2,178 eligible managers.

Survey data were weighted to correct for differences between the sampled individuals and

the population within each city population stratum used in the stratified probability sampling design. The research team calculated sampling weights by comparing the proportion of targeted individuals in each city size grouping within the population to the proportion of sampled individuals from those groupings in the final sample. Table S1.2 presents the rationale and calculation of these sampling weights. The sampling weights were applied in all statistical modeling in this study to ensure representativeness and reduce potential bias.

Table S1.2. Weighting Process

City types	Weighting factor (city population)	# targeted individual in the population	% population	# sampled individual	% sample	Weight (%population / % sample)	# cities in population	# sampled cities	Sampling weight
Small-size (25k-100k)	25-50k	2955	0.498313659	930	0.372	1.33955285	591	186	3.177419
	50-75k	1390	0.234401349	440	0.176	1.33182585	278	88	3.159091
	75-100k	665	0.112141653	210	0.084	1.33501967	133	42	3.166667
Medium-size (100k-250k)	100-125k	340	0.057335582	340	0.136	0.42158516	68	68	1
	125-150k	185	0.031197302	185	0.074	0.42158516	37	37	1
	150-175k	115	0.019392917	115	0.046	0.42158516	23	23	1
	175-200k	140	0.023608769	140	0.056	0.42158516	28	28	1
	200-225k	90	0.015177066	90	0.036	0.42158516	18	18	1
	225-250k	50	0.008431703	50	0.02	0.42158516	10	10	1
Totals		5930	100%	2500	100%		1186	500	

S2. Subsample Bias Test

First, I use a t-test to compare the respondents of the job satisfaction variable with all survey respondents, including nonrespondents. Table S2 shows that the respondents significantly volunteer 42 minutes less ($p < 0.01$) than all survey respondents. There are no significant differences regarding other variables.

Table S2. T-test result comparing respondents of job satisfaction and all survey respondents

Variable	Mean of respondents of job satisfaction (n=187)	Mean of survey respondent (n=621)	Mean Difference	P Value
Volunteer hours	2.6	3.3	-0.7**	0.005
Volunteer diversity	1.4	1.5	-0.1	0.408
Public participation	3.6	3.6	0	0.824
Job satisfaction	4.6	4.6	0	1
Working experience in public sector	24.9	24.3	0.6	0.458
Education	4.6	4.6	0	0.903
Salary	4.3	4.3	0	0.379
Age	52.4	52	0.4	0.484
Gender (1=female, 0 = male)	29.1	28.7	0.4	0.922
Race: white (1=white, 0=other)	89.4	86.7	2.7	0.31
ICT use for public participation	67.9	69.1	-1.2	0.76
Department size	13.4	15	-1.6	0.557
Department types: Mayor's office (1=yes, 0=no)	16	16.1	-0.1	0.984
Department types: Community Development (1=yes, 0=no)	22.5	24.3	-1.8	0.598
Department types: Parks and Recreation (1=yes, 0=no)	18.7	19.5	-0.8	0.815
Department types: Police (1=yes, 0=no)	20.9	19.8	1.1	0.757
Department types: Finance (1=yes, 0=no)	21.9	20.3	1.6	0.635
City-level NGO density	4.4	4.5	-0.1	0.207
Community closeness	0.1	0.1	0	0.200

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

S3. Bivariate Correlation Matrix

Table S3. Bivariate Correlation Matrix

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 Volunteer hours	1.000																	
2 Volunteer diversity	0.399***	1.000																
3 Public participation exposure	0.049	0.056	1.000															
4 Job satisfaction	0.158*	0.112	0.145	1.000														
5 Working experience in public sector	0.012	-0.013	0.053	0.161*	1.000													
6 Education	0.038	0.006	0.146***	0.066	-0.114**	1.000												
7 Salary	0.010	0.031	0.125**	0.267***	0.268***	0.215***	1.000											
8 Age	-0.014	-0.074	0.027	0.126	0.672***	-0.101*	0.197***	1.000										
9 Gender (1=female, 0 = male)	-0.130**	-0.011	0.075	-0.051	-0.152***	0.048	-0.141***	-0.071	1.000									
10 Race: white (1=white, 0=other)	-0.095*	-0.019	0.052	0.034	0.055	-0.078	0.042	-0.052	-0.009	1.000								
11 ICT use for public participation	0.087*	0.141***	0.071	0.194**	0.146***	0.022	0.080	-0.034	-0.051	-0.004	1.000							
12 Department size	0.090*	0.177***	0.016	0.128	0.129**	0.050	0.147***	0.112**	-0.101*	-0.055	-0.019	1.000						
13 Department types: Mayor's office (1=yes, 0=no)	0.010	-0.006	0.147***	0.165*	0.052	0.160***	0.197***	0.019	-0.062	-0.011	0.037	0.138***	1.000					
14 Department types: Community Development (1=yes, 0=no)	-0.004	-0.019	0.204***	-0.149*	-0.099*	0.074	-0.154***	0.024	0.101*	-0.064	-0.141***	-0.119**	-0.248***	1.000				
15 Department types: Parks and Recreation (1=yes, 0=no)	0.045	0.054	0.114**	-0.008	0.041	-0.049	-0.077	-0.008	0.095*	-0.009	0.140***	0.022	-0.218***	-0.282***	1.000			
16 Department types: Police (1=yes, 0=no)	0.114**	0.123**	-0.281***	0.078	0.208***	-0.119**	0.025	-0.036	-0.217***	0.009	0.173***	0.085*	-0.221***	-0.286***	-0.251***	1.000		
17 Department types: Finance (1=yes, 0=no)	-0.166***	-0.153***	-0.189***	-0.070	-0.192***	-0.057	0.034	0.001	0.072	0.079	-0.199***	-0.106**	-0.216***	-0.279***	-0.244***	-0.248***	1.000	
18 City-level NGO density (per 10,000 population)	-0.066	-0.002	0.076	-0.068	0.010	-0.035	-0.116**	-0.013	-0.037	0.063	0.010	0.043	-0.004	0.043	-0.005	-0.017	-0.020	1.000
19 Community closeness	-0.059	-0.048	-0.046	0.026	-0.062	-0.094*	-0.326***	-0.042	0.023	0.080	-0.070	0.030	-0.003	0.005	0.038	-0.048	0.009	0.381***

*Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$. Spearman correlation analysis is used as it does not require variables to conform to a normal distribution and does not require a linear relationship between variables.*

S4. Sources of Dropped Observations

The final analytical sample reflects listwise deletion of observations with missing values on one or more modeled variables. For full transparency, we report the number of dropped observations and associated missing variables across all model specifications.

Table S4. Number of dropped observations and associated missing values

Variables with missing values	Number of observations					
	Model (1)–(2)	Model (3)–(4)	Model (5)– (6)	Model (7)–(8)	Model (9)–(10)	Model (11)–(12)
Missing public participation exposure only	8	9	N/A	N/A	5	5
Missing job satisfaction only	N/A	N/A	387	434	384	430
Missing volunteering hours only	60	N/A	14	N/A	14	N/A
Missing volunteering diversity only	N/A	0	N/A	0	N/A	0
Missing more than one modeled variables	136	179	90	52	92	55
Total observations with missing values	204	188	491	486	495	490

Note: N/A indicates the models do not include such variables

S5. Zero-inflation and Overdispersion

The two indicators of the dependent variable, *volunteering hours* and *volunteering diversity*, are count variables that take nonnegative integer values. They exhibit a right-skewed distribution with excessive zero values (see figure S5). Among the respondents, 22.5% reported volunteering zero hours in a typical week, while 15.5% did not volunteer for any external organization. Most respondents volunteer between zero and two hours in a typical week and engage with zero to three types of organizations. A small proportion of respondents contribute more hours or participate in various organizations. Both *volunteering hours* and *volunteering diversity* show over-dispersion in their distribution. This is evident as the variance of *volunteering hours* (15.8) exceeds its mean (3.3), and the variance of *volunteering diversity* (1.6) exceeds its mean (1.5).

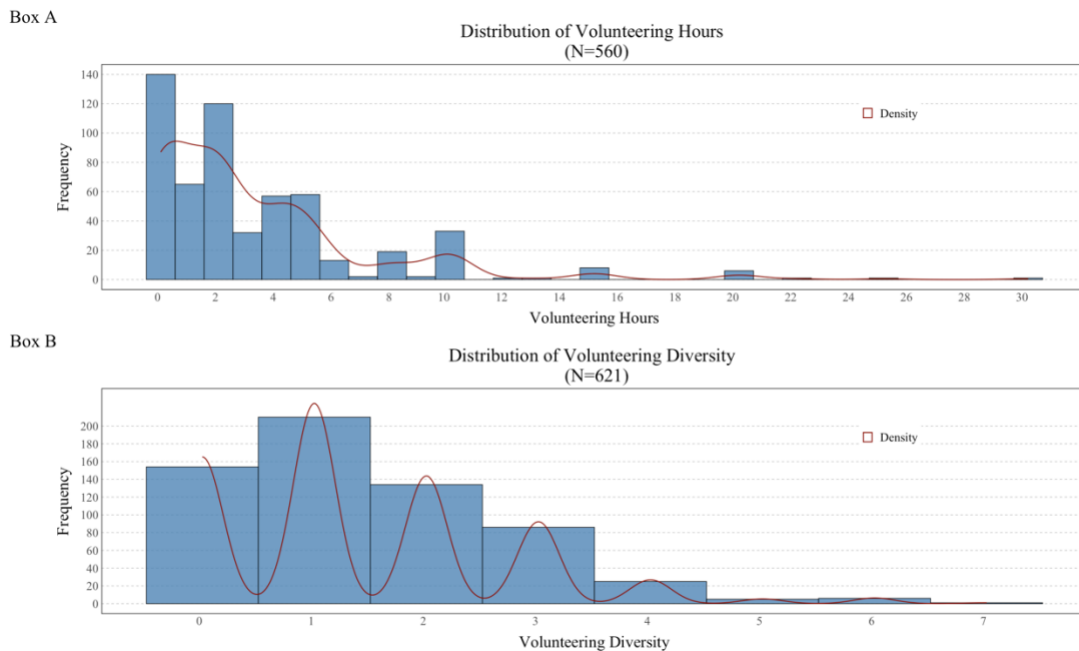


Figure S5. Distribution of dependent variables: volunteering hours and volunteering diversity

S6. Common Method Variance Analysis

This study incorporates several strategies to minimize common method variance (CMV) and social desirability bias (Podsakoff et al. 2003). First, during the survey design phase, survey questions related to public participation, job satisfaction, and volunteering were placed in different sections of the questionnaire. Only job satisfaction items are psychologically perceptual constructs, while the items concerning public participation and volunteering encourage respondents to reflect on the objective reality of their behaviors and organizations. The physical separation of questions makes it unlikely that respondents will connect their answers across these three constructs. Second, survey items relevant to public participation, job satisfaction, and volunteering are well-tested and commonly used in research, ensuring measurement validity. Third, the research team collected the data as an independent, third-party research institution without collaborating with local authorities, promising data confidentiality and respondents' anonymity. Fourth, I include data from other institutional sources, including department types, city population, and city-level NGO density, to reduce bias stemming from CMV.

I conduct Harman's one-factor analysis as a *post hoc* remedy to check CMV. All survey items used in the measurement were included in the test. I used both Principal Components Analysis and Principal Axis Factoring method to detect five latent factors. Results show that the proportion of variance explained by the first latent factor is 16% (see Table S6), which indicates that the factor only explains 23% of the variance in these survey items. Problematic common method variance only occurs when the single factor accounts for more than 50% of the variance among the survey items. The result indicates that no general single factor accounts for 50% or more of the variance among all survey items in the dataset (Fuller et al. 2016). Thus, the test result indicates that common method variance is not a significant concern for this study.

Table S6 Proportion of Variance Explained by Latent Factors

Proportion of variance explained	Latent factor 1	Latent factor 2	Latent factor 3	Latent factor 4	Latent factor 5
Principal Components Analysis	0.23	0.22	0.20	0.19	0.16
Principal Axis Factor	0.35	0.30	0.17	0.10	0.08

Note: Both Principal Components Analysis (PCA) and Principal Axis Factor (PAF) methods were utilized to extract five latent factors. All variables are standardized. The included variables are volunteering hours, volunteering diversity, two items for exposure to public participation, three items for job satisfaction, work experience in public sector, education, salary, age, gender, race: white, department size.

S7. Alternative Public Participation Measure Results

Table S7 Alternative Public Participation Measure Check Results

Outcome variables	Volunteering Hours		Volunteering Diversity	
	(1) 0 or >0	(2) Positive counts	(3) 0 or >0	(4) Positive counts
Public participation exposure	1.15 (0.97-1.13)	0.93 [†] (0.86 – 1.00)	0.94 (0.78–1.14)	1.13** (1.04 – 1.23)
Control variables	Yes		Yes	
R2 (adjusted R2)	0.395 (0.371)		0.063 (0.027)	
Log Likelihood	-2202		-1420	
Observation	417		433	
	Volunteering Hours		Volunteering Diversity	
	(5) 0 or >0	(6) Positive counts	(7) 0 or >0	(8) Positive counts
Public participation exposure	5.39 (0.17 – 174.95)	1.99 (0.57 – 6.89)	1.91 (0.06 – 65.27)	24.54 (1.95 – 308.25)*
Job Satisfaction	15.44 (1.64–163.02)*	1.10 (0.48–2.53)	6.09 (0.63 – 59.06)	7.21 (1.33 – 39.28)*
Public Participation exposure × Job Satisfaction	0.71 (0.33 – 1.53)	0.88 (0.68 – 1.15)	0.75 (0.35 – 1.62)	0.55 (0.32 – 0.94)*
Control variables	Yes		Yes	
R2 (adjusted R2)	0.366 (0.259)		0.144 (0.006)	
Log Likelihood	-553.4		-364	
Observation	126		126	

Note: 95% Confidence Interval in parentheses. [†]p < 0.1, *p < 0.05, **p < 0.01, ***p < 0.001. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

S8. Robustness Check for Model Results without Control Variables

Table S8 Model Results without Control Variables

Outcome variables	Volunteering Hours					
	(1) 0 or >0	(2) Positive counts	(3) 0 or >0	(4) Positive counts	(5) 0 or >0	(6) Positive counts
Public participation exposure	1.22 (1.08 – 1.37)**	0.92 (0.86 – 0.98)**			0.49 (0.07–3.73)	1.72 (0.43–6.85)
Job Satisfaction			3.71 (2.32–5.93)***	0.88 (0.71–1.08)	0.68 (0.15–3.09)	2.19 (0.73–6.53)
Public Participation exposure × Job Satisfaction					1.16 (0.75–1.79)	0.92 (0.69–1.23)
Control variables	No		No		No	
R2 (adjusted R2)	0.003 (-0.000)		0.040 (0.029)		0.016 (-0.006)	
Log Likelihood	-2977		-845.9		-553.4	
Observation	612		173		182	
	Volunteering Diversity					
	(7) 0 or >0	(8) Positive counts	(9) 0 or >0	(10) Positive counts	(11) 0 or >0	(12) Positive counts
Public participation exposure	1.06 (0.95 – 1.19)	1.07 (1.01 – 1.14)*			0.49 (0.07–3.73)	1.72 (0.43–6.85)
Job Satisfaction			1.06 (0.68–1.64)	1.75 (1.31–2.33)***	0.68 (0.15–3.09)	2.19 (0.73–6.53)
Public Participation exposure × Job Satisfaction					1.16 (0.75–1.79)	0.92 (0.69–1.23)
Control variables	No		No		No	
R2 (adjusted R2)	0.003 (-0.000)		0.011 (0.000)		0.016 (-0.006)	
Log Likelihood	-2153		-648.7		-364	
Observation	552		187		182	

Note: 95% Confidence Interval in parentheses. *p < 0.05, **p < 0.01, ***p < 0.001. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

S9. Heckman Selection Model

We conduct a two-stage Heckman selection model to correct for subsample bias (Heckman 1979), following Wooldridge's recommended procedure (Procedure 19.2, 809–812, 2010). Specifically, the two-stage model is structured as follows:

$$Volunteering\ Hours = \beta_1\delta_1 + \alpha_1 JobSatisfaction + \mu_1(1)$$

$$JobSatisfaction = \beta_2\delta_2 + v_2 \quad (2)$$

$$ObservedIndex = 1 [\beta_3\delta_3 + v_3 > 0] \quad (3)$$

Equation (1) is a Hurdle Negative Binomial model predicting volunteering hours based on explanatory and control variables. Equation (2) is a linear regression model predicting job satisfaction using demographic control variables. Equation (3) is the first-stage selection equation, using an observed index coded as "1" if respondents provided answers for both job satisfaction and volunteering hours and "0" if either measures have missing value. This selection model estimates the likelihood of respondents providing both measurements based on demographic variables for the full sample. From this model, I computed the Inverse Mills Ratio and subsequently include it into the outcome model (Equation 4), which is estimated only for respondents with complete observations for both volunteering hours and job satisfaction.

$$Volunteering\ Hours_i = \beta_1\delta_1 + \alpha_1 JobSatisfaction + \gamma_1 InverseMillsRatio + \mu_1(4)$$

Table S9 displays the results of the Heckman selection model. The comparison focuses on the job satisfaction coefficients to evaluate the subsample bias.

Table S9 Heckman Selection Model Results

Outcome variables	Volunteering Hours			
	(1) 0 or >0	(2) Positive counts	(3) 0 or >0	(4) Positive counts
Public participation exposure			2.91 (0.13 – 64.34)	4.90 (1.67–14.35)**
Job Satisfaction	6.56 (3.26–13.22)***	0.74 (0.60–0.90)**	12.71 (1.24 – 130.13)*	2.11 (0.95–4.69)
Public Participation exposure × Job Satisfaction			0.82 (0.42 – 1.62)	0.73 (0.58–0.92)**
Control variables	Yes		Yes	
R2 (adjusted R2)	0.391 (0.299)		0.416 (0.311)	
Log Likelihood	-572		-542.2	
Observation	130		126	

Volunteering Diversity				
	(5) 0 or >0	(6) Positive counts	(7) 0 or >0	(8) Positive counts
Public participation exposure			0.07 (0.00 – 2.10)	11.39 (4.35–29.82)***
Job Satisfaction	3.87 (1.79–8.36)**	1.55 (1.04–2.29)*	0.89 (0.08 – 9.39)	5.61 (2.33–13.49)***
Public Participation exposure × Job Satisfaction			1.66 0.79 – 3.48	0.64 (0.52–0.78)***
Control variables	Yes		Yes	
R2 (adjusted R2)	0.111 (-0.024)		0.148 (-0.005)	
Log Likelihood	-648.7		-345.8	
Observation	130		126	

Note: 95% Confidence Interval in parentheses. †p < 0.1, *p < 0.05, **p < 0.01, ***p < 0.001. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

S10. Alternative Models Diagnostics Comparison

Table S10.1 Models Diagnostics Comparison for Model 1

Model 1	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike's information criterion (AIC)
OLS	5	27.6	33.324	-1166.882	2367.765
Logistics	91	1.516	2.429	-493.582	1019.164
Poisson	22	6.273	9.892	-2884.213	5800.426
NB	96	1.438	2.55	-2248.292	4530.583
Hurdle Poisson	106	1.302	5.014	-2459.788	4983.576
Model 1: Hurdle NB	106	1.302	2.812	-2198.617	4463.234

Note: This table presents a diagnostic comparison of alternative models for Model 1, which examines the association between public participation and volunteering hours using a Hurdle Negative Binomial regression, with additional covariates controlled for. "NB" denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 1 is 138. The expected number of zeros is estimated based on each model's fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostic results show that Model 1: Hurdle NB is preferable.

Table S10.2 Models Diagnostics Comparison for Model 2

Model 2	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike's information criterion (AIC)
OLS	61	2.492	3.019	1419.098	61
Logistics	92	1.652	2.306	883.557	92
Poisson	105	1.448	2.054	2961.782	105
NB	174	0.874	2.054	2963.793	174
Hurdle Poisson	82	1.854	2.611	2907.041	82
Model 2: Hurdle NB	82	1.854	2.583	2908.856	82

Note: This table presents a diagnostic comparison of alternative models for Model 2, which examines the association between public participation and volunteering diversity using a Hurdle Negative Binomial regression, with additional covariates controlled. "NB" denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 2 is 152. The expected number of zeros is estimated based on each model's fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostic results show that Hurdle NB is preferable.

Table S10.3 Models Diagnostics Comparison for Model 3

Model 3	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike's information criterion (AIC)
OLS	7	5.857	14.103	-298.321	630.641
Logistics	28	1.464	2.488	-137.545	307.09
Poisson	15	2.733	5.309	-681.52	1395.04
NB	38	1.079	2.649	-639.289	1312.577
Hurdle Poisson	34	1.206	3.709	-579.416	1222.833
Model 3: Hurdle NB	34	1.206	3.522	-578.575	1223.151

Note: This table presents a diagnostic comparison of alternative models for Model 3, which examines the association between job satisfaction and volunteering hours using a Hurdle Negative Binomial regression, with additional covariates controlled. "NB" denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 3 is 41. The expected number of zeros is estimated based on each model's fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostic results show that Hurdle NB is preferable.

Table S10.4 Models Diagnostics Comparison for Model 4

Model 4 (N=173)	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike's information criterion (AIC)
OLS	24	1.833	2.497	-192.884	419.767
Logistics	29	1.517	2.313	-120.885	273.771
Poisson	40	1.1	1.837	-436.104	904.209
NB	60	0.733	1.837	-436.107	906.214
Hurdle Poisson	29	1.517	2.994	-386.837	837.673
Model 4: Hurdle NB	29	1.517	3.023	-386.837	839.673

Note: This table presents a diagnostic comparison of alternative models for Model 4, which examines the association between job satisfaction and volunteering diversity using a Hurdle Negative Binomial regression, with additional covariates controlled. "NB" denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 3 is 44. The expected number of zeros is estimated based on each model's fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostic results show that Hurdle NB is preferable.

Table S10.5 Models Diagnostics Comparison for Model 5

Model 5 (N=154)	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike's information criterion (AIC)
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OLS	8	5.125	12.962	-281.437	600.874
Logistics	27	1.519	2.666	-135.483	306.966
Poisson	16	2.562	5.054	-645.451	1326.901
NB	38	1.079	2.599	-611.852	1261.705
Hurdle					
Poisson	34	1.206	3.49	-549.229	1170.457
Model 5:					
Hurdle NB	34	1.206	3.459	-549.139	1172.277

Note: This table presents a diagnostic comparison of alternative models for Model 5, which examines the interaction between public participation and job satisfaction and their association with volunteering hours using a Hurdle Negative Binomial regression, with additional covariates controlled for. “NB” denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 5 is 41. The expected number of zeros is estimated based on each model’s fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostic results show that Hurdle NB is preferable.

Table S10.6 Models Diagnostics Comparison for Model 6

Model 6 (N=161)	Expected zero	Ratio (Expected Zero/Observed Zero)	Dispersion	Log- likelihood	Akaike’s information criterion (AIC)
OLS	23	1.913	2.592	-187.228	412.456
Logistics	28	1.571	2.248	-117.482	270.963
Poisson	39	1.128	1.891	-427.427	890.853
NB	58	0.759	1.891	-427.429	892.858
Hurdle					
Poisson	29	1.517	3.112	-370.085	812.169
Model 6:					
Hurdle NB	29	1.517	3.146	-370.085	814.169

Note: This table presents a diagnostic comparison of alternative models for Model 5, which examines the interaction between public participation and job satisfaction and their association with volunteering diversity using a Hurdle Negative Binomial regression, with additional covariates controlled for. “NB” denotes Negative Binomial model. The observed number of zero values in the dependent variable in the analyzed dataset for Model 6 is 44. The expected number of zeros is estimated based on each model’s fitted values, and the ratio represents the expected zeros divided by the observed zeros. Diagnostics results confirm that Hurdle NB is better and desirable.

S11. Robustness Check for Hurdle Negative Binomial Model Results by Comparing to Alternative Model Specifications

Table S11.1 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, there is a significant association between volunteering hours and public participation exposure in hurdle Poisson and original models. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression, while the positive counts part is consistent with the Poisson and negative binomial models.

Table S11.1 Model Specifications on the Association between Public Participation Exposure and Volunteering Hours

N=417	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Public participation exposure	-0.12 (-0.51–0.26)	1.18 (1.01–1.38)*	0.97 (0.93–1.00)	0.96 (0.89–1.04)	1.18 (1.01–1.38)*	0.91 (0.87–0.94)***
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.087/0.053	0.076	0.559	0.178	0.290/0.261	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p < 0.05, **p < 0.01, ***p < 0.001.

“NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

Table S11.2 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, there is a significant association between the intensity of volunteering diversity and public participation exposure in hurdle Poisson and original models. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression, while the positive counts part is consistent with the Poisson and negative binomial models.

Table S11.2 Model Specifications on the Association between Public Participation Exposure and Volunteering Diversity

N=433						
	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Public participation exposure	0.05 (-0.06–0.17)	1.00 (0.84–1.19)	1.04 (0.99–1.10)	1.04 (0.99–1.10)	1.00 (0.84–1.19)	1.09 (1.01–1.18)*
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.099/0.067	0.079	0.22	0.22	00.062 / 0.026	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p < 0.05, **p < 0.01, ***p < 0.001. “NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

Table S11.3 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, there is a significant association between volunteering hours and job satisfaction in hurdle Poisson and original models, compared to OLS. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression, while the positive counts part is opposite to the Poisson and negative binomial models. This indicates that the coefficients in the Poisson and negative binomial models are overestimated and driven by the zero-inflation.

Table S11.3 Model Specifications on the Association between Job Satisfaction and Volunteering Hours

N=130						
	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Job Satisfaction	0.40 (-0.54–1.34)	5.59 (2.94–11.19)***	1.17 (0.99–1.38)	1.29 (1.01–1.65)*	5.59 (2.87–10.89)***	0.75 (0.63–0.90)**
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.215/0.111	0.219	0.726	0.480	0.364 / 0.274	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p<0.05, **p<0.01, ***p<0.001. “NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

Table S11.4 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, there is a significant association between volunteering diversity and job satisfaction in hurdle Poisson and original models, compared to OLS. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression, while the positive counts part is consistent with the Poisson and negative binomial models.

Table S11.4 Model Specifications on Effect of Job Satisfaction on Volunteering Diversity

N=135						
	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Job Satisfaction	0.32 (-0.07–0.70)	2.49 (1.25–5.07)*	1.32 (1.06–1.66)*	1.32 (1.06–1.66)*	2.49 (1.24–4.98)*	1.42 (0.99–2.04)†
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.220/0.121	0.251	0.419	0.419	0.123 / 0.004	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p < 0.05, **p < 0.01, ***p < 0.001.

“NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

Table S11.5 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, job satisfaction has a significant moderating effect on the association between public participation and the intensity of volunteering hours in both the hurdle Poisson and original models, compared to OLS. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression, while the positive counts part is consistent with the Poisson and negative binomial models.

Table S11.5 Model Specifications on the Moderating Effect of Job Satisfaction on the Association between Public Participation Exposure and Volunteering Hours

N=126						
	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Public participation exposure	1.65 (-2.61–5.92)	1.39 (0.07–31.30)	2.48 (1.06–5.90)*	3.96 (1.15–14.03)*	1.39 (0.07, 29.56)	4.85 (1.71–13.74)**
Job satisfaction	1.43 (-1.65–4.52)	6.35 (0.71–65.62)	2.12 (1.15–3.99)*	3.27 (1.31–8.45)**	6.35 (0.67, 60.01)	2.11 (0.97–4.58)
Public participation exposure × Job satisfaction	-0.29 (-1.21–0.63)	0.97 (0.49–1.88)	0.85 (0.70–1.01)	0.76 (0.58–0.99)*	0.97 (0.50, 1.89)	0.73 (0.59–0.91)**
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.216/0.093	0.220	0.712	0.500	0.390/0.288	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p < 0.05, **p < 0.01, ***p < 0.001. “NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.

Table S11.6 indicates that coefficients from original models (i.e., hurdle negative binomial) are consistent with hurdle Poisson models. After accounting for zero-inflation and overdispersion, job satisfaction has a significant moderating effect on the association between public participation and the intensity of volunteering diversity in hurdle Poisson and original models. The coefficients from the zero-inflated part of the original models (0 or >0) are consistent with the logistic regression. The statistical significance of the positive counts part is higher than that of the Poisson and negative binomial models, because the Poisson and negative binomial models are affected by zero-inflation.

Table S11.6 Model Specifications on the Moderating Effect of Job Satisfaction on the Association between Public Participation Exposure and Volunteering Diversity

N=126						
	OLS	Logistic	Poisson	NB	H Poisson	
					0 or >0	Positive counts
Public participation exposure	0.13 (-1.75–2.01)	0.31 (0.01, 7.61)	1.54 (0.49–5.06)	1.54 (0.49–5.06)	0.31 (0.01, 7.61)	11.69 (1.22–111.89)*
Job satisfaction	0.34 (-1.02–1.70)	1.58 (0.17, 15.08)	1.69 (0.72–4.09)	1.69 (0.72–4.09)	1.58 (0.17, 15.08)	5.42 (1.03–28.49)*
Public participation exposure × Job satisfaction	-0.02 (-0.42–0.39)	1.19 (0.60, 2.37)	0.92 (0.72–1.18)	0.92 (0.72–1.18)	1.19 (0.60, 2.37)	0.63 (0.39–1.01)†
Control variable	Yes	Yes	Yes	Yes	Yes	
R ² / Adj R ²	0.216/0.093	0.220	0.712	0.500	0.390/0.288	

Note: 95% Confidence Interval in parentheses. †p<0.1, *p < 0.05, **p < 0.01, ***p < 0.001.

“NB” denotes Negative Binomial model. “H” denotes Hurdle. To ensure comparability of coefficients across models, the incidence rate ratio is reported as a coefficient for each model.